

Understanding Generator Droop Voltage Control

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Abstract—Power park modules are required by network codes to provide droop voltage power control at the point of connection (POC) to the grid. This paper contains a thorough analysis of the droop voltage control scheme of power park modules by comparing it with the usual terminal voltage control scheme of synchronous generators in small-test cases. Moreover, it explains the impact of generators under droop voltage control on the power flow solution.

Keywords—Power park module, Droop voltage control, Power flow solution.

I. INTRODUCTION

Most synchronous generators are connected to the grid through their step-up transformer. The Automatic Voltage Regulator (AVR) usually controls the generator terminal voltage[1]. The AVR implements either a high gain proportional or integral control.

Voltage-reactive power control of power park modules (i.e. wind farms and solar photovoltaic plants) has evolved from power factor control to voltage control. Recent network codes have required power park modules to provide droop voltage control at the point of connection (POC) to the grid [2]. The POC is typically at the high-voltage side of the step-up transformer of the power park module. As power park modules are based on distributed resources (wind generators or solar photovoltaic inverters), the power plant controller is in charge of implementing such a control scheme while the distributed resources are controlled as reactive power sources.

Investigating the impact of the droop voltage control of power park modules on power system steady-state performance requires incorporating such a control scheme in the power flow model as it has been explained in [3]. However, reference [3] does not provide any insight on the behavior of the droop voltage control. Moreover, the impact of substituting synchronous generators by power park modules on power system steady-state performance must be analyzed to understand its implications for a successful energy transition ([4], [5]). Hence, this paper explains the impact of generators under droop voltage control on the power flow solution.

The rest of the paper is organized as follows. Section II . formulates the droop voltage control. Section III details two models to help understanding the integral control of synchronous generator terminal voltage and the droop control of the POC voltage of power park modules. Section IV explains the meaning of the power flow solution with generator droop voltage control and provides two illustrative

examples of different sizes. Finally, Section V provides the conclusions of the paper.

II. DROOP VOLTAGE CONTROL

A droop voltage control is a proportional control in which the generator reactive power variation is proportional to its voltage variation (see Figure 1).

$$\Delta q_G = -K \Delta v \quad (1)$$

The gain K is equal to the inverse of the voltage-reactive power droop R . The droop is in the range between 2 and 7% in per unit of the maximum reactive power supplied by the generator and in per unit of the voltage.

$$K = \frac{1}{R} = \frac{\Delta q (pu)}{\Delta v (pu)} = \frac{\Delta Q/Q_{max}}{\Delta V/V_B} \quad (2)$$

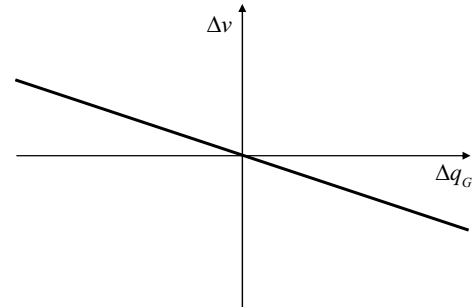


Figure 1. Linearized model of a droop voltage control.

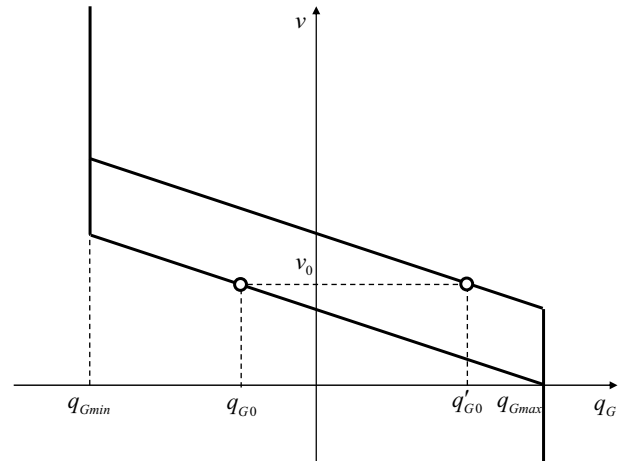


Figure 2. Non-linear model of a droop voltage control.

Droop voltage control is similar to droop frequency control when the linearized model is considered. However, the differences between droop voltage and frequency control become more apparent if the non-linear model is considered (see Figure 2 and Figure 3).

$$q_G - q_{G0} = -K (v - v_0) \quad (3)$$

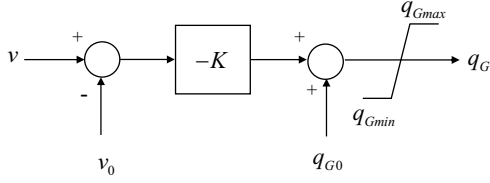


Figure 3. Non-linear model block diagram.

Whereas the setpoint frequency, f_0 , of the frequency control is uniquely defined, the voltage setpoint, v_0 , of the voltage control is not uniquely defined and depends on the objective and solutions of a superior optimal power flow. In other words, the implementation of a droop voltage control requires the specification of a pair of voltage and reactive power (q_{G0}, v_0) . If only the voltage is specified, the voltage-reactive power curve is not unique. Figure 2 shows two different voltage-reactive power curves that would yield to two different reactive power values (q_{G0} and q'_{G0}) for the same voltage setpoint.

III. SIMULATION OF GENERATOR DROOP VOLTAGE CONTROL

Two models and the associated test cases have been developed to help understanding the integral control of synchronous generator terminal voltage and the droop control of POC voltage of power park modules. The relationship between the gain of the droop voltage control and the reactance of the generator step-up transformer is developed.

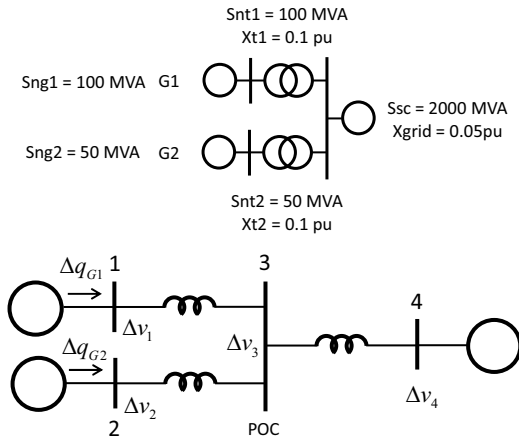


Figure 4. Test case of two synchronous generators with integral terminal voltage control.

Figure 4 shows a test case that consists of two synchronous generators connected through their step-up transformer to a grid represented by its Thévenin equivalent in terms of its short circuit capacity at the POC. The rating of generator G2 is half of the rating of generator G1. The rating of the step-up transformers is identical to the rating of the corresponding generators. The reactances of the step-up transformers are identical in their respective apparent power base. Generator

AVR controls generator terminal voltage and determines generator reactive power generation.

A simple time domain simulation model has been developed in which the relationship between voltage variation and supplied reactive power is governed by the voltage-reactive power equations of the fast-decoupled power flow.

$$\begin{bmatrix} \Delta q_{G1} \\ \Delta q_{G2} \\ 0 \\ \Delta q_{G4} \end{bmatrix} = \begin{bmatrix} b''_{11} & b''_{12} & b''_{13} & b''_{14} \\ b''_{21} & b''_{22} & b''_{23} & b''_{24} \\ b''_{31} & b''_{32} & b''_{33} & b''_{34} \\ b''_{41} & b''_{42} & b''_{43} & b''_{44} \end{bmatrix} \begin{bmatrix} \Delta v_1 \\ \Delta v_2 \\ \Delta v_3 \\ \Delta v_4 \end{bmatrix} \quad (4)$$

Figure 5 displays the block diagram of the time domain simulation model of two synchronous generators with integral terminal voltage control in the case of a step of the external grid voltage Δv_4 . The integral gain of the regulators is equal to 10 in their maximum reactive power base.

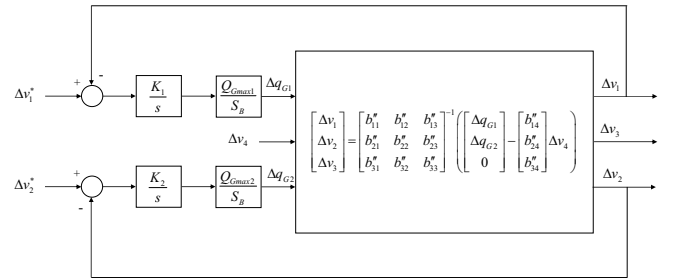


Figure 5. Time domain simulation model of two synchronous generators with integral terminal voltage control.

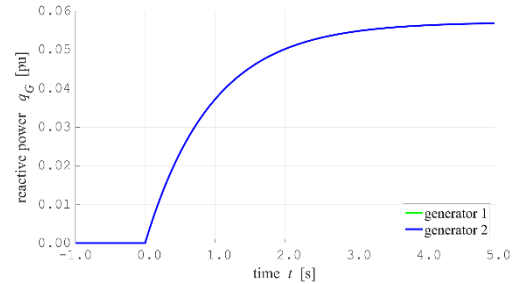


Figure 6. Simulation of two synchronous generators with integral terminal voltage control: Generator reactive powers.

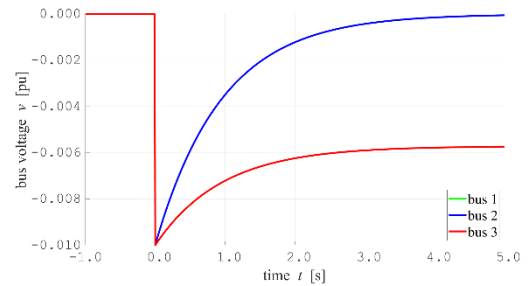


Figure 7. Simulation of two synchronous generators with integral terminal voltage control: Bus voltages.

Figure 6 shows the reactive power supplied by each generator in their apparent power base. They are identical. Hence, reactive power generation is proportional to the size of the generator provided that the transformer reactances in each generator apparent power base are identical. Figure 7 shows the bus voltages. Both generator terminal voltages (bus 1 and bus 2) are identical. The steady-state error of the generator terminal voltages is zero due to the action of the integral

regulators. However, the response of the POC voltage (bus 3) exhibits the characteristic response of a proportional (droop) voltage control.

Figure 8 shows a test case that consists of two power park modules with proportional POC voltage control connected to a grid represented by its Thévenin equivalent in terms of its short circuit capacity at the POC. Note that the step-up transformer is behind the POC (within the power park module). The rating of power park module G2 is half of the rating of power park module G1. Power park module voltage control determines power park reactive power generation following the droop voltage control.

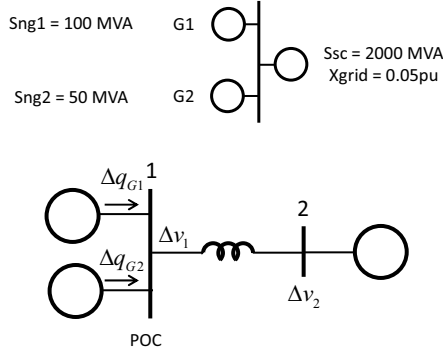


Figure 8. Test case of two power parks with proportional PCC voltage control.

A simple time domain simulation model has been developed in which the relationship between voltage variation and supplied reactive power is governed by the voltage-reactive power equations of the fast-decoupled power flow

$$\begin{bmatrix} \Delta q_{G1} + \Delta q_{G2} \\ 0 \end{bmatrix} = \begin{bmatrix} b''_{11} & b''_{12} \\ b''_{21} & b''_{22} \end{bmatrix} \begin{bmatrix} \Delta v_1 \\ \Delta v_2 \end{bmatrix} \quad (5)$$

Figure 9 displays the block diagram of the time domain simulation model of two power park modules with droop control of POC voltage in the case of a step of the external grid voltage Δv_2 . Please note that it has been assumed without loss of generality that the gain of the droop voltage control is 33.33 in the respective apparent power bases, which corresponds to a droop of 3%. The maximum reactive power supplied by each generator is 30% of their apparent power base. The time constant of reactive power control loop of the power park modules is 500 ms.

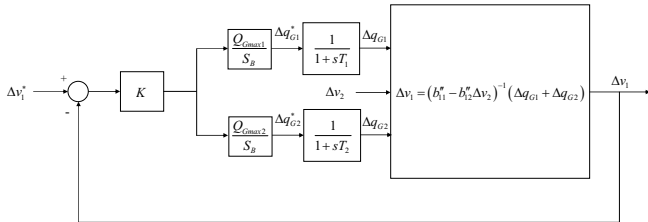


Figure 9. Simulation model of two power park modules with proportional POC voltage control.

Figure 10 shows the reactive power supplied by each generator in their own apparent power base. They are identical as expected. Figure 11 shows the POC voltage. The response of the POC voltage exhibits the characteristic response of a proportional (droop) voltage control as expected.

The performed analysis allows us to conclude that integral control of the synchronous generator terminal voltage is

equivalent to a droop control of the POC voltage of a power park module. Moreover, the gain of the droop control of the POC voltage fulfills:

$$K \cdot q_{Gmax} = \frac{1}{X_t} \quad (6)$$

where X_t is the per unit reactance of the step-up transformer of the synchronous generator and q_{Gmax} is the maximum reactive power in per unit in the same bases.

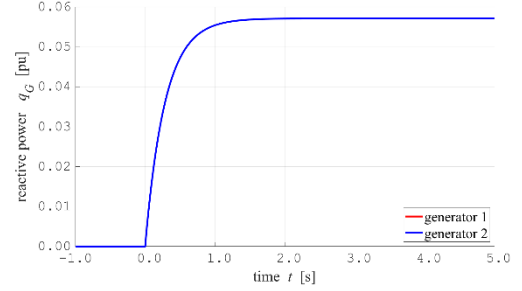


Figure 10. Simulation of two power parks with proportional POC voltage control: Power park reactive powers.

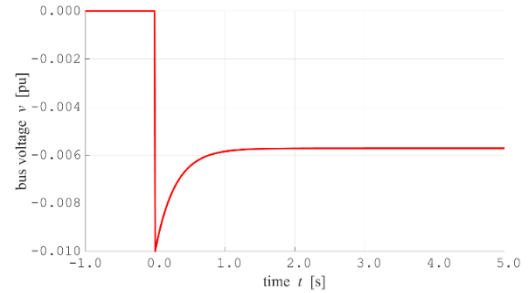


Figure 11. Simulation of two power parks with proportional POC voltage control: PCC voltage.

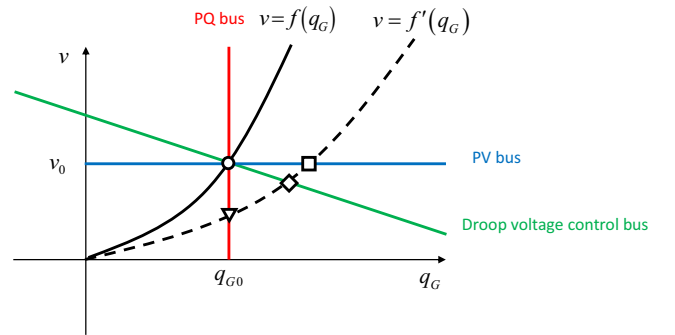


Figure 12. Impact of the generator control scheme on a post disturbance operating point.

IV. POWER FLOW SOLUTION WITH GENERATOR DROOP VOLTAGE CONTROL

Let us consider an operating point in the $q_G - v$ plane. A voltage variation due to the variation of supplied reactive power is described by $v = f(q_G)$. Figure 12 explains the impact of the generator control scheme on a post-disturbance operating point. After a disturbance $v = f(q_G)$ becomes $v = f'(q_G)$. If the bus is a PV one (blue line), the voltage remains constant whereas supplied reactive power increases. In contrast, if the bus is a PQ bus (red line), reactive power remains constant whereas voltage decreases. If the bus is a generator droop voltage control (green line), both voltage, and

reactive power changes. It is important to emphasize what has already been stated in Section II: the implementation of a voltage droop control requires the specification of an operating point (q_{G0}, v_0) no matter how the system has arrived at it.

Two examples are proposed to illustrate the performance of the power flow solution with generator droop voltage control and to understand the impact of generator droop voltage control on power system steady-state performance.

A. Three-bus test system

The single line diagram and the input data of a three-bus test system are displayed in Figure 13. The original operating point assumes that the two circuits between bus #1 and bus #2 are in service. A disturbed operating point arises when one of the two circuits between bus #1 and bus #2 trips. Figure 14 displays the original operating point of bus #2 (with a circle), and the operating point reached after the considered disturbance under different voltage control schemes of bus #2: PV bus (with a square), PQ (with a triangle) bus and generator droop control (with a diamond). Figure 14 confirms the ideas presented in Figure 12.

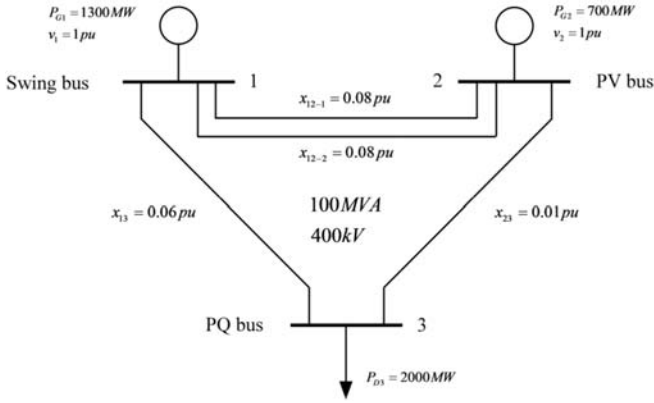


Figure 13. Three-bus test system.

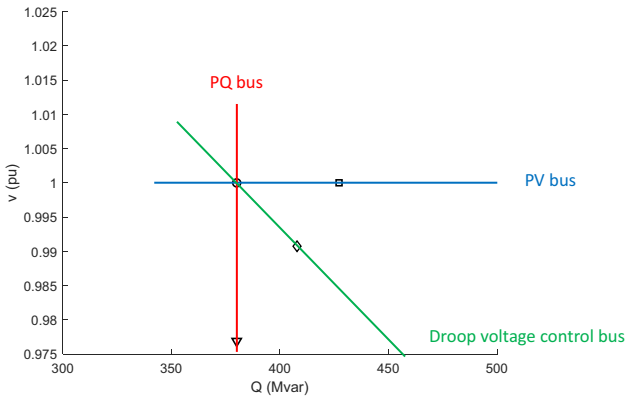


Figure 14. Three bus test system: original operating point and operating points after one circuit tripping from bus #1 to bus #2 under different voltage control schemes.

B. Thirty-nine bus test system

The equivalence of the integral terminal voltage control and the droop control is checked in the thirty-nine-bus test system. Two cases are compared when the line between bus #15 and bus #16 trips: integral terminal voltage control of all generators, and droop control of the POC voltages at bus #19, bus #20, bus #22, and bus #23 with the remaining generators applying an integral terminal voltage control. Figure 15 shows

the single-line diagram of the thirty-nine-bus test system. Figure 16 shows the single-line diagram of the modified thirty-nine-bus test system with droop control of the POC voltages at bus #19, bus #20, bus #22, and bus #23. It actually contains 35 buses since the step-up transformers of generators under droop voltage control have been removed. The generators subject to discussion have been highlighted in red in both Figure 15 and Figure 16.

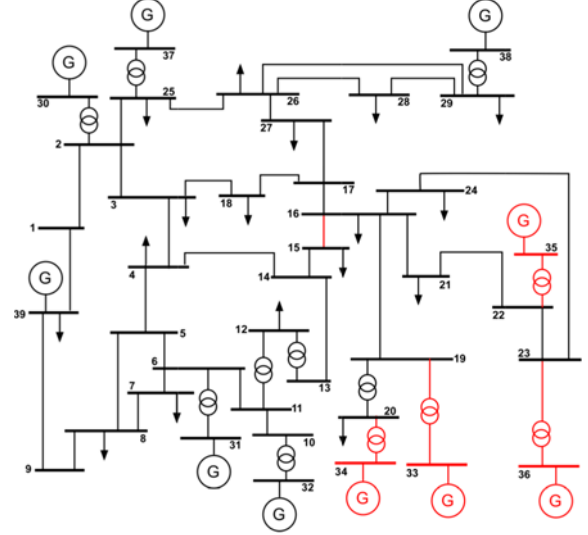


Figure 15. Thirty-nine-bus test system.

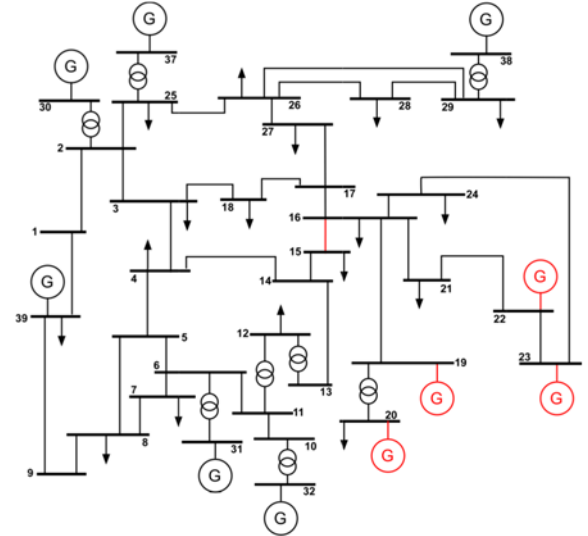


Figure 16. Modified thirty-nine-bus test system:

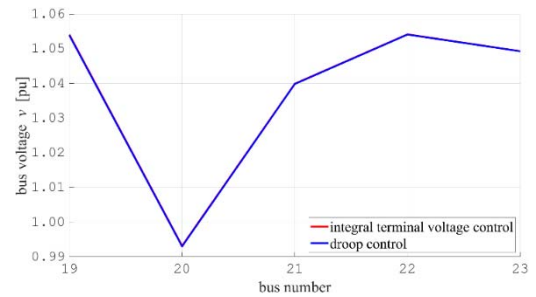


Figure 17. Bus voltages comparison.

Figure 17 and Figure 18 compare respectively the bus voltages and the supplied reactive powers in the area of interest in the two cases considered assuming that the gain of the droop voltage control has been chosen according to (6). As expected, the bus voltages and the supplied reactive powers in

the two cases are almost the same, which confirms the proper performance of the power flow solution with generator droop voltage control.

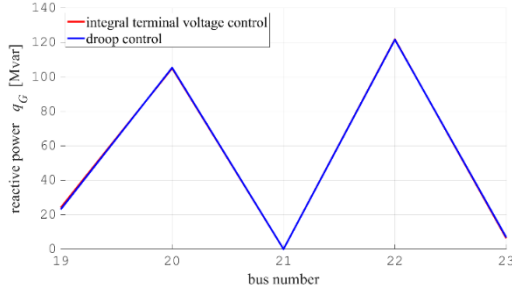


Figure 18. Supplied reactive power comparison.

Further confirmation of the proper performance of the power flow solution with generator droop voltage control can be obtained using the simulation model of Figure 19 in which the output of the droop voltage controllers sets the supplied reactive power of the generators, which are represented as PQ buses. Figure 20 provides a qualitative outline of the time evolution in the voltage-supplied reactive power plane. At t_{0-} the system is operating at a point in which $v_0 = f(q_{G0})$. At t_{0+} the bus voltage droops because of the disturbance $v = f'(q_G)$ since the generator is operating as a PQ bus. The steady-state post-disturbance operating point is reached at t_{∞} by time domain simulation which involves in each time step a power solution in which generators under droop control are represented as PQ buses.

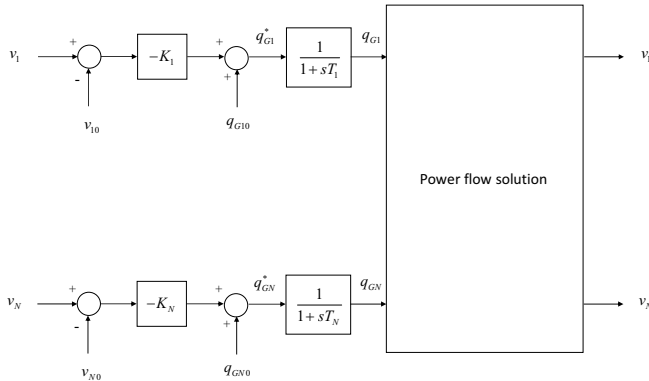


Figure 19. Simulation model with droop POC voltage control.

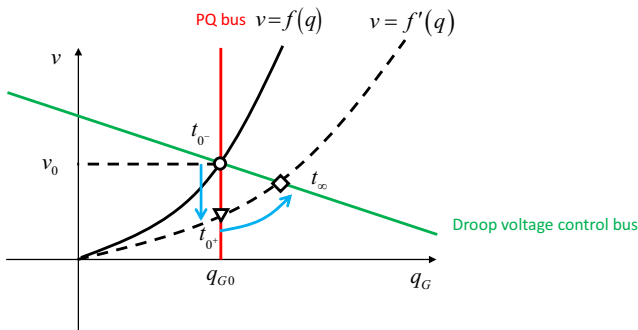


Figure 20. Outline of time response evolution in the voltage-supplied reactive power plane.

Figure 21 and Figure 22 show, respectively, the time variation of the bus voltages and supplied reactive power of the generators under droop control. They confirm the results provided by the power flow solution. It must be noted that

although we assumed that a line tripping would result in a voltage drop, in this specific case, it results in a voltage rise.

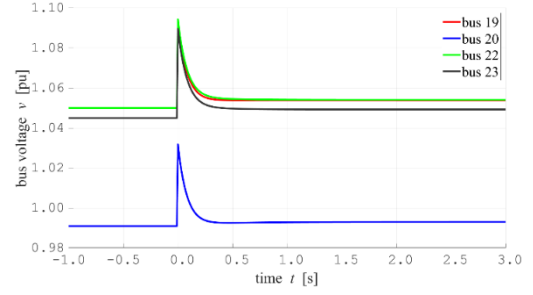


Figure 21. Bus voltages time variation.

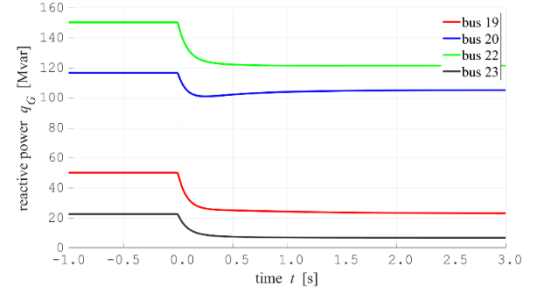


Figure 22. Generator reactive power time variation.

V. CONCLUSIONS

This paper has detailed how to incorporate generators under droop voltage control in the power flow model. In addition, it has provided a thorough analysis of the droop voltage control of power park modules by comparing it with the usual terminal voltage control scheme of synchronous generators. The analysis performed has allowed us to conclude that integral control of synchronous generator terminal voltage is equivalent to power park module droop control of the POC voltage if the droop gain is selected according to the step-up transformer reactance.

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